🔑 **Answer Key — Teacher Copy · Lesson 8-2**

MSTAR PRACTICE WORKSHEET

Lesson 8-2: Write and Solve Equations Using Addition or Subtraction

Grade 6 Mathematics · Standard 6.AT.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Solve: x + 9 = 15

- **A** x = 6
- **B** x = 9
- **C** x = 15
- **D** x = 24

Subtract 9 from both sides: x = 15 − 9 = 6. Check: 6 + 9 = 15 ✓

Question 2 SELECTED RESPONSE — CORRECT: C

Solve: y − 7 = 20

- **A** y = 7
- **B** y = 13
- **C** y = 27
- **D** y = 140

Add 7 to both sides: y = 20 + 7 = 27. Check: 27 − 7 = 20 ✓

Question 3 SELECTED RESPONSE — CORRECT: B

Solve: n + 15 = 42

A n = 15

B n = 27

C n = 42

D n = 57

Subtract 15 from both sides: n = 42 − 15 = 27. Check: 27 + 15 = 42 ✓

Question 4 SELECTED RESPONSE — CORRECT: C

Solve: m − 11 = 25

A m = 14

B m = 25

C m = 36

D m = 275

Add 11 to both sides: m = 25 + 11 = 36. Check: 36 − 11 = 25 ✓

Question 5 SELECTED RESPONSE — CORRECT: D

A shirt costs $d. After a $15 discount, it costs $38. What is d?

A $23

B $38

C $45

D $53

d − 15 = 38 → d = 38 + 15 = 53.

Question 6 SELECTED RESPONSE — CORRECT: B

Unlock Card 1: Solve x + 7 = 15.

A. x = 7

B. x = 8

C. x = 22

D. x = 105

Subtract 7 from both sides: x = 15 − 7 = 8. Check: 8 + 7 = 15 ✓

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: D

Which value of n is the solution of n − 18 = 27?

A. n = 9

B. n = 18

C. n = 27

D. n = 45

The variable has 18 subtracted from it, so add 18 to both sides to isolate n: n = 27 + 18 = 45. Check by substituting: 45 − 18 = 27.

- **A:** That subtracts 18 from 27. The variable already had 18 taken away, so ADD it back.
- **B:** That is the amount subtracted, not the solution. Add it to 27.
- **C:** That is the value after 18 was subtracted, not n itself. Add 18 to recover n.

Part B — correct answer: A

Which reasoning justifies the step used to find that answer?

A. Because 18 is subtracted from n, adding 18 to both sides is the inverse operation that leaves n by itself.

B. Because the equation shows subtraction, 18 should be subtracted from both sides as well.

C. Because the number written after the equal sign is always the value of the variable.

D. Because the smaller number in a subtraction equation is always the solution.

Subtraction is undone by addition. Adding 18 to both sides keeps the equation balanced and isolates the variable, giving n = 45. Subtracting 18 from both sides instead would give 27 − 18 = 9, and 9 − 18 does not equal 27.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Adding When the Inverse Is Subtraction

> Darius solved the equation x + 23 = 40. He wrote: 'x + 23 = 40, so x = 40 + 23, which means x = 63.'

Explain Darius's misconception and give the correct value of x.

Model answer: x = 17. The 23 is being added to x, so the inverse operation is subtraction: subtract 23 from both sides to get x = 40 − 23 = 17. Check by substituting: 17 + 23 = 40. Darius's answer fails the same check, because 63 + 23 = 86, not 40.

2 points	Student explains that adding 23 is undone by subtracting 23 from both sides, not by adding again, and gives x = 17.
1 point	Student gives x = 17 but does not explain why subtraction is the inverse operation needed.
0 points	Student agrees with Darius or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A detective solved x + 25 = 60 and got x = 35. Another detective solved y − 25 = 60 and got y = 35. Are both correct? Explain using inverse operations and check each answer.

Model answer: Only the first detective is right. For x + 25 = 60 the inverse is subtraction, so x = 60 − 25 = 35, and 35 + 25 = 60 checks out. For y − 25 = 60 the inverse is addition, so y = 60 + 25 = 85, not 35 — substituting 35 gives 35 − 25 = 10, not 60. Opposite operations need opposite inverses, even when the numbers look identical.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.